07-26 第2篇

阅读理解（说明文） How Does a Rainbow Form?

① After a summer rain, if the sun comes out quickly, you may look up and find a soft curve of colour stretching across the sky: red on the outside, violet on the inside, with orange, yellow, green, blue, and indigo in between. People have watched rainbows for thousands of years. The ancient Greeks thought rainbows were a bridge for a goddess. But modern science tells us a simpler, equally beautiful story. A rainbow is made in one place only: **inside tiny drops of water, when sunlight bends through them in just the right way.**

② To understand the rainbow, we first need to know that **white sunlight is not really white**. In 1666, a young scientist named Isaac Newton did a famous experiment. He let sunlight pass through a piece of shaped glass called a **prism**. To his surprise, a band of colours came out of the other side — red, orange, yellow, green, blue, indigo, violet. He understood that white light is a mix of all these colours, travelling together. Something in the glass had separated them.

③ What the glass did was to **slow the light down** and **bend it**. Light travels very fast in air but a little slower through glass, water, or any transparent object. When a ray of light enters a piece of glass at an angle, it bends. This bending is called **refraction**. Different colours bend by slightly different amounts — violet bends a little more than red — so a single beam of white light becomes a fan of colours.

④ A raindrop in the sky is a very small, round piece of water — a natural "prism" in the shape of a ball. When sunlight hits the front surface of a raindrop, it bends and separates into its colours. Some of the light then hits the back wall of the drop and bounces back — this is called **reflection**. Finally, the light leaves the drop, bending a second time as it goes back into the air. The result: a small rainbow comes out of every single raindrop, with red on top and violet on the bottom.

⑤ Why, then, do we see the rainbow as a giant curve in the sky? Because the light comes out of each drop at a special angle — about **42°** from a straight line between the sun and the drop. Only the drops that happen to sit at exactly this angle from our eyes send their colours to us. Drops above that angle send colours somewhere else; drops below do the same. The drops that lie at 42° form a big circle around the point directly opposite the sun. We usually see only the top half of this circle, because the ground blocks the rest. From a plane or a tall mountain, a full, round rainbow is sometimes visible.

⑥ The rainbow also explains why we must stand with the **sun behind us** and the **rain in front of us** to see one. Without both, there is no rainbow. And because the angle to our eyes depends on where we stand, every person actually sees a *slightly different* rainbow. Two people walking side by side watch their own private rainbows, made from their own raindrops. If you walk towards a rainbow, it seems to move away — because different drops are now at the 42° angle to you. You can chase a rainbow, but you cannot catch one.

⑦ Sometimes, a faint second rainbow appears above the main one, with its colours in the **opposite order** (violet on top, red on the bottom). This "double rainbow" happens when some of the light reflects **twice** inside the raindrop before escaping. The extra reflection flips the order of the colours and makes the second rainbow a little dimmer.

⑧ So next time a rainbow appears, take a quiet moment to look. What you see in the sky is actually a small crowd of raindrops, each one acting as a tiny glass ball, each one catching the sun, bending its colours, and sending them — along one precise line — straight to your eyes. The rainbow is not above the Earth; it is being made between you and the rain, every moment it shines.

根据文章内容，选择最佳答案。

1. What is the real nature of white sunlight, according to the passage?

A. It is always red at sunrise and yellow at noon.
B. It is a mix of different colours (red, orange, yellow, green, blue, indigo, violet) travelling together.
C. It contains only red and blue.
D. It is made of soft clouds.

2. What did Isaac Newton discover by using a prism in 1666?

A. That the prism was made of rainbow glass.
B. That a prism can turn water into colour.
C. That white light splits into a band of colours because the glass bends different colours by slightly different amounts.
D. That light travels faster in glass than in air.

3. Inside a raindrop, what happens to sunlight to produce a rainbow?

A. The drop absorbs all the colours.
B. Light enters the drop, bends (refraction), bounces off the back wall (reflection), and bends again as it leaves, separating into colours.
C. The drop colours the light with tiny paint.
D. The drop heats the light until it changes colour.

4. Why do we often see only the top half of a rainbow?

A. Because the sun is always directly in front of us.
B. Because the ground blocks the part of the full circle that is below the horizon.
C. Because half the colours are invisible to humans.
D. Because rainbows are always broken in the middle.

5. What is the main message of the last paragraph?

A. A rainbow is far away in space and has nothing to do with us.
B. A rainbow is actually made between the raindrops in front of us and our own eyes; each person sees a rainbow of his or her own.
C. Rainbows are caused by the moon.
D. Only expensive cameras can see rainbows.

答案与解析

1.
【答案】B
【题型】细节理解题
【定位】第②段 "white light is a mix of all these colours..."
【解析】白光是七色混合。A/C/D 均错。故选 B。

2.
【答案】C
【题型】细节理解题
【定位】第②—③段 "Different colours bend by slightly different amounts..."
【解析】折射量不同导致色散。A/B/D 均错。故选 C。

3.
【答案】B
【题型】细节理解题
【定位】第④段 "refraction... reflection... a small rainbow comes out of every single raindrop..."
【解析】折射+内反射+再折射。A/C/D 均错。故选 B。

4.
【答案】B
【题型】细节理解题
【定位】第⑤段 "the ground blocks the rest."
【解析】圆的下半被地面挡住。A/C/D 均错。故选 B。

5.
【答案】B
【题型】主旨大意题

【定位】第⑧段
【解析】彩虹其实是在我们与雨滴之间生成，每人所见独一。A/C/D 均偏。故选 B。

词汇注释（超纲词）

- **violet / indigo** [n.] 紫 / 靛
- **prism** [n.] 棱镜
- **refraction** [n.] 折射
- **reflection** [n.] 反射
- **transparent** [adj.] 透明的